AIN4101: Machine Learning II

Lecture 1: Introduction to Natural Language Processing

Badr University in Assiut (BUA)
School of Artificial Intelligence & Data Management

Week 1, Spring 2026

Outline

1. Course Overview
2. NLP Motivation and Applications
3. Language and Linguistic Levels
4. The NLP Pipeline
5. Tokens and Tokenization
6. From Text to Numerical Representation
7. Summary

Table of Contents

1. Course Overview
2. NLP Motivation and Applications
3. Language and Linguistic Levels
4. The NLP Pipeline
5. Tokens and Tokenization
6. From Text to Numerical Representation
7. Summary

Current Section

Course Overview

Welcome to Machine Learning II!

Course Information

- **Course Code:** AIN4101
- **Credits:** 3 (2 lecture + 1 lab)
- **Prerequisites:** Machine Learning, Neural Networks, Python programming
- **Main Textbook:** Geron, *Hands-On Machine Learning with Scikit-Learn and PyTorch*, O'Reilly, 2025

What You Will Learn

- Modern NLP pipelines and sequence modeling
- RNNs, LSTMs, GRUs, attention, and transformers
- Vision and multimodal transformers
- Autoencoders, generative models, ML design patterns, and MLOps

Course Structure

Lectures (10 weeks):

1. Introduction to NLP
2. Text Representation and Word Embeddings
3. RNNs, LSTM, and GRU for Sequential Data
4. Encoder-Decoder Models and Attention
5. Transformers and Self-Attention
6. Pretrained Language Models and Modern NLP
7. Vision and Multimodal Transformers
8. Autoencoders and Modern Generative Models
9. Machine Learning Design Patterns
10. MLOps and Production Machine Learning

Assessment:

- Lab Assignments & Quizzes: 15%
- Final Project: 10%
- Midterm Exam: 25%
- Final Exam: 50%

Important Dates:

- Midterm: Week 6
- Final Project: Week 10

Learning Objectives - Today

By the end of this lecture, you will be able to:

1. Explain what Natural Language Processing is and why it matters.
2. Identify common NLP applications and distinguish NLU from NLG.
3. Recognize linguistic levels that affect NLP systems.
4. Describe the main steps of a basic NLP pipeline.
5. Define corpus, document, sentence, token, and vocabulary.
6. Explain tokenization and compare word, character, and subword tokenization.
7. Describe how text begins to become numerical input for machine learning models.

Key Takeaway

NLP starts before modeling: we must first decide how language will be segmented, normalized, represented, and connected to a learning algorithm.

Table of Contents

1. Course Overview
2. NLP Motivation and Applications
3. Language and Linguistic Levels
4. The NLP Pipeline
5. Tokens and Tokenization
6. From Text to Numerical Representation
7. Summary

Current Section

NLP Motivation and Applications

What is Natural Language?

Natural Language

Natural language is the language people use to communicate with each other in speech, writing, and conversation.

- It includes words, grammar, meaning, context, and social use.
- It can appear as text, speech, dialogue, documents, web pages, messages, or instructions.
- It is flexible, ambiguous, and strongly dependent on context.

Example

The word *bank* can refer to a financial institution or the side of a river. The intended meaning depends on context.

What is Natural Language Processing?

Definition

Natural Language Processing (NLP) is a field of artificial intelligence and computational linguistics concerned with enabling computers to process, analyze, understand, and generate human language.

- **Process:** split, normalize, and structure language data.
- **Analyze:** identify patterns, topics, entities, sentiment, or intent.
- **Understand:** infer meaning from language in context.
- **Generate:** produce text, summaries, answers, or dialogue.

Why is NLP Important?

Human Side

- Humans communicate knowledge through language.
- Large parts of digital information are unstructured text.
- Users often prefer asking questions in natural language.

Machine Side

- ML algorithms need numerical representations.
- Text is variable-length and symbolic.
- NLP transforms language into data that models can learn from.

Important Note

The central difficulty is not only building a model; it is also deciding how raw language becomes structured numerical input.

Common Applications of NLP

Application	Typical Goal
Machine Translation	Translate text or speech from one language to another.
Information Retrieval	Find relevant documents or passages for a user query.
Information Extraction	Extract entities, dates, relations, events, or facts from text.
Sentiment Analysis	Classify opinions or emotions in text.
Question Answering	Produce or retrieve answers to natural-language questions.
Chatbots	Carry out task-oriented or open-ended dialogue.
Summarization	Compress long documents into shorter summaries.

Natural Language Understanding vs. Generation

Natural Language Understanding (NLU)

- Input: language
- Output: meaning, label, structure, or decision
- Examples: intent detection, sentiment analysis, entity recognition, question understanding

Natural Language Generation (NLG)

- Input: data, prompt, representation, or context
- Output: coherent language
- Examples: text generation, summarization, dialogue response, explanation generation

Key Takeaway

Many modern systems combine both: they interpret an input, reason over context, then generate a response.

Table of Contents

1. Course Overview
2. NLP Motivation and Applications
3. Language and Linguistic Levels
4. The NLP Pipeline
5. Tokens and Tokenization
6. From Text to Numerical Representation
7. Summary

Current Section

Language and Linguistic Levels

Why is Language Difficult for Machines?

- **Ambiguity:** one expression can have multiple meanings.
- **Context dependence:** meaning can depend on previous sentences, speaker intention, or world knowledge.
- **Variation:** spelling, dialect, slang, abbreviations, typos, and domain-specific terminology vary widely.
- **Compositionality:** word meanings combine through grammar to form sentence meanings.
- **Open vocabulary:** new names, products, places, and technical terms appear continuously.

Example

"You have a green light" can mean a physical green light, permission to cross, or approval to continue a plan.

Linguistics and Language

Linguistics

Linguistics is the scientific study of language: its sounds, words, grammar, meaning, and use.

Why it matters for NLP

- NLP systems model language as data, but language has internal structure.
- Linguistic structure helps us think clearly about words, sentences, meaning, and discourse.
- Modern NLP often learns representations automatically, but many errors still reflect linguistic difficulty.

Levels of Language Analysis

Level	Main Question
Lexical	What are the words or tokens?
Morphological	What meaningful parts form each word?
Syntactic	How are words arranged grammatically?
Semantic	What is the literal meaning?
Pragmatic	What does the speaker intend in context?
Discourse	How do utterances connect across a conversation or document?

Important Note

These levels are not isolated in real systems. A model may need information from several levels at once.

Lexical and Morphological Analysis

Lexical Analysis

- Focuses on words or tokens in text.
- Often begins with segmenting text into smaller units.

Example

I love programming $\rightarrow$ [I, love, programming]

Morphology

- Studies meaningful components of words.
- Morphemes can mark tense, number, derivation, or grammatical role.

Example

unhappiness $\rightarrow$ un $+$ happy $+$ ness

Syntactic Analysis

Syntax

Syntax studies how words are arranged into grammatical phrases and sentences.

Grammatical

John eats an apple.

Ungrammatical

Apple eats John an.

- Syntax affects meaning and model predictions.
- Word order is especially important in sequence models and transformers.
- Later lectures model word order using RNNs, attention, and self-attention.

Semantics and Pragmatics

Semantics

- Concerns literal meaning.
- Asks what entities, actions, attributes, and relations are expressed.

Pragmatics

- Concerns language use in context.
- Asks what the speaker intends and how the listener should react.

Example

"Sara opened the door" expresses an action, an actor, and an object.

Example

"Can you open the window?" is usually a request, not a question about ability.

Discourse

Discourse

Discourse concerns linguistic units larger than one sentence, such as paragraphs, documents, conversations, and dialogue turns.

- Sentences depend on what came before them.
- Pronouns require previous context.
- Conversation systems must track dialogue state and user goals.
- Retrieval and summarization require document-level organization.

Example

"Mary bought a laptop. She uses it every day." The words *she* and *it* depend on the previous sentence.

Table of Contents

1. Course Overview
2. NLP Motivation and Applications
3. Language and Linguistic Levels
4. The NLP Pipeline
5. Tokens and Tokenization
6. From Text to Numerical Representation
7. Summary

Current Section

The NLP Pipeline

A Basic NLP Pipeline

$$\text{Raw Text} \rightarrow \text{Preprocessing} \rightarrow \text{Tokenization} \rightarrow \text{Vocabulary/IDs} \rightarrow \text{Representation} \rightarrow \text{Model} \rightarrow \text{Prediction}$$

Pipeline Idea

- NLP models do not directly consume raw human language.
- Text is prepared, segmented, mapped to IDs, and represented numerically.
- The representation is then passed to a machine learning or deep learning model.

Corpus, Document, Sentence, and Token

Term	Meaning in NLP
Corpus	A collection of texts used for analysis or model training.
Document	One text item inside a corpus, such as an article, review, message, or paragraph.
Sentence	A linguistic unit often used for segmentation and analysis.
Token	A unit produced by tokenization, such as a word, character, punctuation mark, or subword.
Vocabulary	The set of token types known to the model or tokenizer.

Text Preprocessing: Purpose and Caution

Purpose

Text preprocessing prepares raw text for computational processing while preserving information needed for the target task.

- Preprocessing choices depend on the task and model type.
- Lowercasing may help in some tasks but may remove useful information in others.
- Removing punctuation may hurt tasks where punctuation carries meaning.
- Removing stop words is not always appropriate, especially for modern neural models.

Important Note

Preprocessing is not a fixed checklist. Each step should be justified by the task, data, and model.

Common Preprocessing Operations

Typical Operations

- Unicode decoding
- Case normalization
- Sentence segmentation
- Tokenization
- Handling punctuation
- Handling URLs, emails, emojis, or numbers

Classical NLP Operations

- Stop-word filtering
- Stemming
- Lemmatization
- N-gram extraction
- Sparse feature extraction

Sentence Segmentation

Definition

Sentence segmentation is the task of breaking text into sentence-like units.

- It is often needed before sentence-level classification, translation, or summarization.
- It is not always trivial: periods may appear in abbreviations, initials, URLs, and numbers.
- Modern NLP pipelines often use pretrained tokenizers or rule-based components to handle these cases.

Example

"Dr. Ali arrived at 9 a.m. He gave a lecture." contains periods that are not all sentence boundaries.

Table of Contents

1. Course Overview
2. NLP Motivation and Applications
3. Language and Linguistic Levels
4. The NLP Pipeline
5. Tokens and Tokenization
6. From Text to Numerical Representation
7. Summary

Current Section

Tokens and Tokenization

What is Tokenization?

Tokenization

Tokenization is the process of separating running text into units called tokens.

- Tokens can be words, punctuation marks, characters, or subword units.
- Tokenization is often one of the first steps in modern NLP systems.
- The choice of tokenization affects vocabulary size, unknown words, and model input length.

Example

I want to learn NLP. $\rightarrow$ [I, want, to, learn, NLP, .]

Word-Level Tokenization

Idea

Split text into words and punctuation, then map each word type to an ID.

- Simple and intuitive for many languages.
- Can produce very large vocabularies.
- Struggles with rare words, new words, spelling variation, and morphologically rich languages.
- May be difficult for languages without explicit word boundaries.

Example

“playing” and “played” may become separate tokens even though they are related.

Character-Level Tokenization

Idea

Represent text as a sequence of characters.

- Very small vocabulary.
- Can represent unseen words using characters.
- Sequences become much longer than word-level sequences.
- The model must learn word-like and morpheme-like patterns from characters.

Example

NLP $\rightarrow$ [N, L, P]

Subword Tokenization

Idea

Subword tokenization represents text using units between characters and full words.

- Frequent words may remain as one token.
- Rare words can be decomposed into smaller pieces.
- The vocabulary is smaller than word-level vocabulary but sequences are shorter than character-level sequences.
- This approach is widely used in modern pretrained language models.

Example

unhappiness $\rightarrow$ [un, happi, ness] or similar subword pieces, depending on the tokenizer.

Byte Pair Encoding: Intuitive Idea

BPE Intuition

Byte Pair Encoding starts from small units and repeatedly merges frequent adjacent pairs to create larger subword tokens.

1. Start with characters or byte-level units.
2. Count frequent neighboring pairs in a corpus.
3. Merge the most frequent pair.
4. Repeat until the desired vocabulary size is reached.

Key Takeaway

BPE helps represent rare or unseen words using multiple subword tokens instead of a single unknown token.

Special Tokens

What are Special Tokens?

Special tokens are symbols added by tokenizers or models to mark structure in the input.

Token	Typical Use
`[CLS]`	Classification or sequence-level representation in some encoder models.
`[SEP]`	Separator between segments or sentences.
`[PAD]`	Padding shorter sequences to the same length in a batch.
`[UNK]`	Represents unknown or out-of-vocabulary tokens.
`<s>, </s>`	Beginning and end markers in some models.

Padding, Truncation, and Attention Masks

Padding

Adds `[PAD]` tokens so sequences in a batch have the same length.

Truncation

Cuts long sequences to fit the model's maximum input length.

Attention Mask

Marks which positions are real tokens and which are padding.

Example

Input IDs: [101, 2023, 2003, 102, 0, 0] Mask: [1, 1, 1, 1, 0, 0]

Table of Contents

1. Course Overview
2. NLP Motivation and Applications
3. Language and Linguistic Levels
4. The NLP Pipeline
5. Tokens and Tokenization
6. From Text to Numerical Representation
7. Summary

Current Section

From Text to Numerical Representation

Vocabulary and Token IDs

Vocabulary

A vocabulary is the set of token types that a tokenizer or model can map to integer IDs.

- Machine learning models do not operate directly on strings.
- Tokenizers map each token to an integer ID.
- The same sentence becomes a sequence of IDs.

Example

Tokens	I	like	NLP
IDs	12	305	981

One-Hot and Sparse Representations

One-Hot Encoding

A token can be represented as a vector whose length equals the vocabulary size, with one entry equal to 1 and all other entries equal to 0.

- Simple but very sparse.
- Does not directly encode semantic similarity.
- Leads naturally to classical text representations such as Bag of Words and TF-IDF.

Example

If vocabulary = [AI, ML, NLP], then NLP $\rightarrow$ [0, 0, 1].

Embeddings: Dense Representations

Embedding

An embedding maps a token ID to a dense vector of learned numerical features.

- Dense vectors are shorter than one-hot vectors.
- They are learned from data or reused from pretrained models.
- Similar tokens may have similar vectors depending on the training objective.
- Embeddings are the bridge from token IDs to neural networks.

$$\text{Token ID} \rightarrow \text{Embedding Lookup} \rightarrow \text{Dense Vector} \rightarrow \text{Neural Model}$$

A First NLP Classification Pipeline

Reviews $\rightarrow$ Tokenization $\rightarrow$ Numerical Representation $\rightarrow$ Classifier $\rightarrow$ Sentiment Label

Example Task: Sentiment Analysis

- Input: "The movie was surprisingly good."
- Representation: token IDs, Bag of Words, TF-IDF, or embeddings
- Model: logistic regression, RNN, transformer, or another classifier
- Output: positive / negative / neutral

Today's Key Concepts

1. **NLP** enables computers to process, analyze, understand, and generate human language.
2. **Applications** include translation, retrieval, extraction, sentiment analysis, question answering, chatbots, and summarization.
3. **Language levels** include lexical, morphological, syntactic, semantic, pragmatic, and discourse analysis.
4. **Tokenization** converts running text into tokens.
5. **Subword tokenization** balances vocabulary size and ability to handle rare words.
6. **Numerical representation** converts text into forms that machine learning models can process.

Key Takeaway

Lecture 1 answers: What is NLP, why is it useful, and how does raw text begin to become model input?

Next Lecture Preview

Lecture 2: Text Representation and Word Embeddings

- Bag of Words
- N-grams
- TF-IDF
- Sparse vs. dense representations
- Word embeddings
- Embedding layers
- Semantic similarity

Lab 1

- Load a small text dataset
- Apply sentence segmentation and tokenization
- Build a vocabulary
- Convert text into token IDs and sparse features

Recommended reading: Jurafsky & Martin Ch. 1-2; Geron Ch. 14; scikit-learn; Hugging Face Tokenizers.

Questions?

Additional Resources

- Jurafsky & Martin, *Speech and Language Processing*, 3rd ed. draft, 2026.
- Geron, *Hands-On Machine Learning with Scikit-Learn and PyTorch*, O'Reilly, 2025.
- Hugging Face Tokenizers documentation.
- scikit-learn text feature extraction documentation.